Kai Tang Sundararaj

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EDUCATION

University of Waterloo | Waterloo, Ontario
M.Eng. in Computer and Electrical Engineering
Graduating May 2028

Carleton University | Ottawa, Ontario
B.S. in Computer Science, Honours | GPA: 4.0/4.0
Relevant Courses: Data Structures/Algorithms, Web Applications, Systems Programming, Database Management

AWARDS AND EXTRACURRICULARS

University Medal: Awarded for graduating in the top 1% of my class at Carleton
Carleton University Top Entrance Scholarship: Highest level of entrance scholarship
John R. Pugh Scholarship of Excellence: For being the top student in my year within the computer science department
Varsity Badminton Team: Play men's doubles for Carleton's varsity badminton team.
Carleton Dean's Honour list: 2021-2022, 2022-2023, 2023-2024, 2024-2025

PROFESSIONAL EXPERIENCE

Software Engineering Intern | Ericsson **Winter 2025, Summer 2025, Fall 2025**
- Refactored product code in C to improve code health, and improved efficiency by up to 100%
- Debugged failing test cases and implemented fixes
- Used a virtual machine, Linux, Gerrit, Jenkins, and Git
- Led a team of interns to identify and optimize areas of product code
- Participated in daily stand-up meetings as part of Agile Development

Software Developer Intern | Crypto4a Technologies **Fall 2023**
- Used Open API, Maven, MongoDB, and Java to write a REST API inbox service and the backend code behind it for the cloud team where users could send, receive, accept, decline, query, and set expiry dates for invitations to other users.
- Wrote tests for MongoDB storage drivers as well as ledger and inbox services using Junit and Mockito.

PROJECTS

Graph Algorithm Visualizer | Java, Spring Boot, JavaScript, HTML, CSS | Live Demo **March - June 2025**
- Created a site where users can draw their own mazes/graphs on a grid and visualize graph algorithms such as Dijkstra's, Astar, DFS, and BFS in motion on a graph where the nodes can have different edge costs.
- Used Java Spring Boot backend server that takes the input from the frontend, applies delays, and solves the path using an algorithm while sending updates as the graph is solved from starting point to finish point using a WebSocket.
- Calculated number of nodes traversed and the value of the path found to compare algorithms.

Boggle Solver | Java, Spring Boot, JavaScript, HTML, CSS | Live Demo **May - August 2025**
- Developed a site that can solve any boggle grid.
- Users can enter their own boggle grid from or have the site generate a random one.
- Once they submit their boggle the backend will solve it using a prefix tree and DFS, and once it is solved it will send all the words back to the frontend where the user can hover or click them to see the path for that word.

TECHNICAL SKILLS

- Programming Languages: Java, Go, C, C++, Python, JavaScript, TypeScript, HTML/CSS, Haskell, Prolog
- Other Technologies: Git, GitHub, Linux, Spring Boot, Node.js, Express, React, SQL, Postgres, MongoDB, Jira, Jenkins, Gerrit
- Interests: Badminton, Piano, Reading, Video Games, Board Games
- Languages: EN, FR, Conversational Mandarin